

AI Comprehension Evaluation

Paid per-school add-on · Pricing model, unit economics & sales pitch summary

■ The pitch

“Every child reads aloud to their teacher every night – without costing the teacher a minute of listening.”

- **The problem it solves:** comprehension recordings already exist in Lumi and are gold – but unlistenable at scale. 25 kids × 1 min/night is 2+ hours per week per class, so in practice nobody listens and the evidence dies after 90 days.
- **Morning triage:** by the next school morning, a glanceable class view – who understood what they read, who's coasting on minutes, and which three children need a check-in *today*. Intervention lands weeks earlier than a term assessment would catch it.
- **Comprehension, not just minutes:** Lumi already proves children *read*; this proves they *understood* – mapped to literacy dimensions teachers actually use (literal recall → sequencing → inference → vocabulary → personal connection).
- **Termly evidence on tap:** a one-tap “Term 1 Reading Report – Johnny Smith”: reading stats plus comprehension growth with evidence lines, edited and approved by the teacher. Hours saved per report cycle, stronger parent-teacher conversations, whole-school literacy data for leadership.
- **Teacher in control:** the AI never speaks to parents or children; every judgement sits one tap from the original audio; “verify before acting” disclaimers on every surface. An assistant, not an assessor.
- **Privacy-first (the objection-killer):** children's voice audio never leaves Australia (Google Cloud, Sydney region). Only transcript text reaches the evaluation model – no student identifiers are attached and the student's registered name is redacted before sending (spoken content may incidentally mention names; the collection notice covers this). School-level opt-in, defined retention and deletion windows, no training on school data.
- **Principal ROI math:** at A\$12/student/yr this buys the equivalent of ~1,000 teacher-hours of listening coverage (300 kids × 200 nights × 1 min) – roughly the cost of one workbook per child, for a nightly window into every child's comprehension.

What we never promise: not a safeguarding or monitoring system (content flags are best-effort triage only) · not a formal assessment instrument (teacher judgement is the assessment) · partial-term data when enabled mid-term · results land “by next school morning”, not in real time.

■ Unit economics – what one recording costs us

≈ 2.6¢

AUD COST PER RECORDING (45S)

A\$5.20

COGS / STUDENT / YR @ 100%
PARTICIPATION

A\$2.60-3.10

COGS / STUDENT / YR @ REALISTIC
50-60%

PIPELINE STAGE	BASIS	PER RECORDING (USD)	PER RECORDING (AUD)
Google Speech-to-Text v2 (Sydney region)	45s @ ~US\$0.016/min	\$0.0120	1.8¢
Claude Haiku 4.5 evaluation	~2,000 tokens in @ \$1/M + ~500 out @ \$5/M	\$0.0045	0.7¢

Cloud Functions / Firestore / storage	amortised	\$0.0005	0.1¢
Total marginal cost per recording		≈ \$0.017	≈ 2.6¢

FX assumed at AUD/USD 0.66. Question-classification calls are cached per unique question and amortise to ~zero. Costs are bounded per school by a provisioned daily cap and metered per school per month for invoice reconciliation.

■ Worked example – the reference school

300 students • 45-second recording each school night (Mon-Fri) • 200 school days/year – i.e. the theoretical maximum where every child records every night.

LINE	VOLUME / YEAR	ANNUAL (USD)	ANNUAL (AUD)
Recordings	300 × 200 = 60,000	–	–
Speech-to-Text	45,000 audio-minutes	\$720	A\$1,090
Claude Haiku 4.5	60,000 evaluations	\$270	A\$410
Infra (functions, Firestore, storage)	–	~\$40	~A\$60
COGS @ 100% participation		≈ \$1,030	≈ A\$1,560

PARTICIPATION SENSITIVITY	RECORDINGS / YR	ANNUAL COGS (AUD)	PER STUDENT / YR
40% of school nights	24,000	A\$625	A\$2.10
60% (expected realistic band)	36,000	A\$935	A\$3.10
80%	48,000	A\$1,250	A\$4.15
100% (contract worst case = the cap)	60,000	A\$1,560	A\$5.20

■ **Suggested list pricing (AUD ex GST, annual)**

Feature is enabled per school by the Lumi team once commercials are agreed (super-admin switch – no self-service). Billed via the existing per-student line-item invoice; per-school usage metering reconciles every invoice against actual consumption.

SCHOOL SIZE	LIST PRICE	FLOOR	300-STUDENT EXAMPLE	GROSS MARGIN (WORST / REALISTIC)
Up to 150 students	A\$15 / student / yr	A\$1,200	–	–
151 – 400 students	A\$12 / student / yr	–	A\$3,600 / yr (A\$300 / mo)	~57% / ~75%
401 – 800 students	A\$10 / student / yr	–	–	improves with scale
800+ students	A\$8 / student / yr or custom	–	–	–

- **Alternate framings:** A\$4/student/term (aligns with the term-report cadence) · free first term for pilot schools, convert at term end with their own data as the proof.
- **Margin floor is contractual:** the per-school daily cap means a school can never consume more than ~its student count in evaluations per day – worst-case COGS is bounded by design, so margin can't be blown out by usage.

■ **Fleet economics at scale**

FLEET (300-STUDENT SCHOOLS)	ANNUAL REVENUE @ A\$12	ANNUAL COGS (REALISTIC-MAX)	GROSS PROFIT
10 schools	A\$36,000	A\$9k – 16k	A\$20k – 27k
50 schools	A\$180,000	A\$47k – 78k	A\$102k – 133k
200 schools	A\$720,000	A\$187k – 312k	A\$408k – 533k

Unit costs improve at scale: Google STT volume tiering, Anthropic Batch API (-50%) for overnight processing, and prompt/token tuning are all untapped levers in these numbers.

■ **Scaling & cost-control posture (how the margin is protected)**

- **Freshness SLA:** “results by next school morning.” Recordings cluster 6–8pm; the pipeline drains the evening peak overnight. Throughput is a config knob (function instance count), not an architecture change – headroom to ~50,000 recordings/day.
- **Four independent spend guards:** platform kill switch (fail-closed – if config is unreadable, nothing spends) · per-school daily cap sized to enrolment at provisioning · global daily cap · Anthropic console spend limit + GCP budget alerts.
- **Nothing is dropped at cap:** over-cap recordings defer to the next day's budget automatically.
- **Per-school metering:** monthly usage & cost counters per school power invoice reconciliation and margin monitoring in the super-admin dashboard.